

MICROALERT:
DRIVER MICROSLEEP DETECTION APP



Abstract

Real-time microsleep detection for driver safety using EAR



Uses smartphone camera for facial and eye monitoring



Detects microsleep and triggers instant sound & visual alerts



Designed for practical, safe use by Grab drivers

Objective

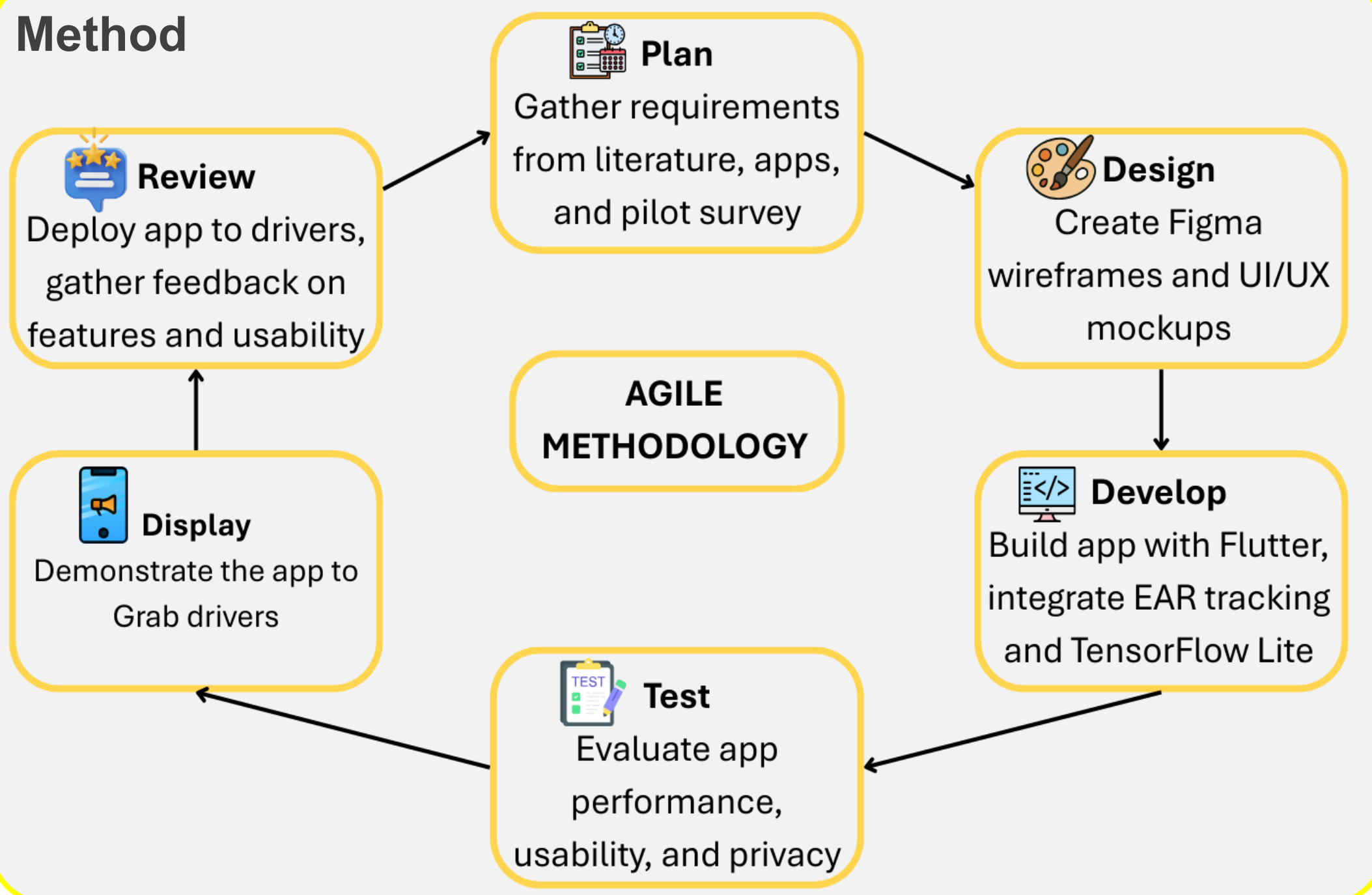
1. To **study** driver microsleep issues by reviewing **current detection apps**, identifying user needs **surveys**, and examining technological limitations via **literature reviews**.
2. To **develop** a mobile app for **microsleep detection**.
3. To **assess** the **usability** of the developed application through **User Acceptance Testing (UAT)** and user feedback.



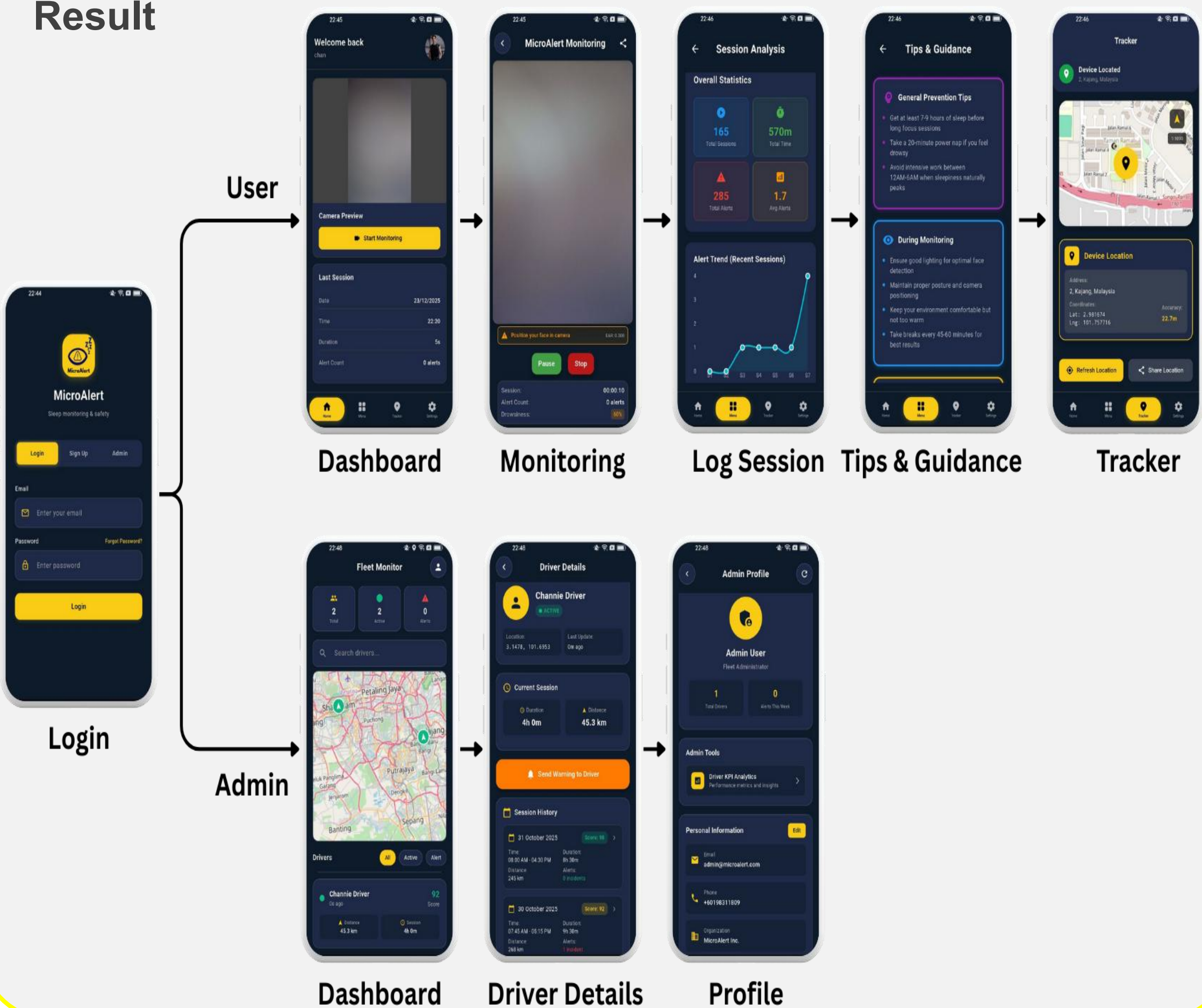
Comparative Study

Feature	Literature Review	Existing App	Survey Findings	MicroAlert
Detection Method	CNN/LSTM and EEG are accurate but heavy. EAR is suitable for mobile	RoadCat & Awaker: camera-based eye closure PREDICT: wearables	100% want real-time facial monitoring	Real-time facial and EAR-based detection on mobile
Alert Mechanism	Audio vibration and visual alerts effective if timely	RoadCat uses sound only. Awaker uses sound and visual. PREDICT uses wearables.	Prefer sound and visual with customization	Sound, vibration and visual alerts with adjustable sensitivity
Adaptability to Environment	Lighting and head pose affect accuracy	RoadCat and Awaker sensitive to lighting. PREDICT needs wearables	Drivers operate day and night	Designed to work across varied lighting conditions
Accessibility & Availability	Mobile vision is most practical	Awaker not in Malaysia. PREDICT requires hardware	Low awareness. High willingness to adopt	Smartphone-based and suitable for Malaysian drivers

Method



Result



Result & Conclusion

